

IT3091 – Bank Marketing Evaluation Log

Step	Findings	Evidence
1. Initial Section	XGBoost selected as leading candidate based on PR-AUC	PR-AUC: XGBoost 0.4210 vs Random Forest 0.4098
2. Threshold selection performed	Business-cost framework (illustrative cost/call = \$5, value/subscriber = \$80) applied to find the profit-optimal threshold for each model	XGBoost optimal profit \$44,750 at threshold 0.31; Random Forest optimal profit \$46,455 at threshold 0.35
3. Discrepancy noticed	Random Forest achieves ~3.8% higher expected profit despite lower PR-AUC	Profit gap: \$1,705 in favor of Random Forest
4. Statistical validation	Bootstrap resampling (500 resamples, seed 42) confirms the gap is not sampling noise	Mean (RF – XGBoost) profit difference: \$1,785; 95% CI [\$282, \$3,465]; Random Forest superior in 99.2% of resamples
5. Mechanistic explanation	Precision-recall curves cross at approximately recall = 0.66. XGBoost leads below this point (high-precision, low-recall region); Random Forest leads above it	XGBoost profit-optimal recall = 76.7%; Random Forest profit-optimal recall = 81.0% - both past the crossover, in Random Forest's stronger region
6. Conclusion	PR-AUC is a global ranking metric averaged across all thresholds; it does not guarantee superiority at the <i>specific</i> threshold a business will actually deploy. Random Forest is superior in exactly the operating region this task requires	Final model selection reversed: Random Forest , not XGBoost

Limitation: The bootstrap captures test-set sampling variability only (what if a different 9,043 clients had been drawn), not training variability (a different random seed or resampled training set). Cost/value figures (\$5/\$80) are illustrative, not confirmed bank figures — see the separate sensitivity analysis for how the recommended threshold shifts under different assumptions.

01. Candidates carried forward

Random Forest and XGBoost, per Model Comparison Log conclusion. Retrained on the final feature set(bank_train_final.csv/bank._test_final, 40 features, contact_unknown removed), random_state = 42, in a separate Evaluation notebook.

02. Precision-Recall curve comparison

Observations	Details
Both models rank meaningfully better than random	Both curves lie well above the base rate (11.7%) across nearly all recall levels
High-confidence region	At low recall (0.0–0.2), both models achieve ~0.70–0.75 precision — highest-scored clients are genuinely most likely to subscribe
Initial visual read	XGBoost appeared to dominate across recall 0.3–0.8; Random Forest started slightly higher at very low recall but dropped off faster
Convergence	Both curves converge toward the base rate as recall → 1.0 (expected — full recall means calling everyone)
Overall summary metric	PR-AUC: XGBoost 0.4210 vs Random Forest 0.4098
Later finding (see section 9)	The "XGBoost dominates 0.3–0.8" read was imprecise — the actual crossover is at recall ≈ 0.66 , and the profit-optimal operating points for both models sit past that crossover, in Random Forest's stronger region

03. Confusion matrices across candidate thresholds (XGBoost)

Threshold	TN	FP	FN	TP
0.3	4615	3370	237	821
0.4	6030	1955	358	700
0.5	6919	1066	447	611
0.6	7353	632	536	522
0.7	7571	414	623	435

04. Threshold table — business interpretation (XGBoost)

Threshold	Calls Made	% Clients Called	Subscribers Caught	Subscribers Missed	Wasted Calls	Precision	Recall
0.1	7908	87.4%	1017	41	6891	0.1286	0.9612
0.2	6143	67.9%	926	132	5217	0.1507	0.8752
0.3	4191	46.3%	821	237	3370	0.1959	0.7760
0.4	2655	29.4%	700	358	1955	0.2637	0.6616
0.5	1677	18.5%	611	447	1066	0.3643	0.5775
0.6	1154	12.8%	522	536	632	0.4523	0.4934
0.7	849	9.4%	435	623	414	0.5124	0.4112

0.8	567	6.3%	326	732	241	0.5750	0.3081
0.9	269	3.0%	177	881	92	0.6580	0.1673

05. Cost-based threshold optimization

Since the dataset provides no real cost/revenue figures, an illustrative business-cost framework was used: cost per call = \$5, value per subscriber = \$80 (breakeven marginal precision = $5/80 = 6.25\%$). For XGBoost, expected net profit peaks at **threshold ≈ 0.31 , profit $\approx \$44,750$** . Marginal analysis confirms this is a genuine optimum, not an artifact: moving from threshold $0.4 \rightarrow 0.3$ adds clients at marginal precision $\approx 7.9\%$ (above the 6.25% breakeven, so profit rises); moving from $0.3 \rightarrow 0.2$ adds clients at marginal precision $\approx 5.4\%$ (below breakeven, so profit falls).

06. Threshold sensitivity analysis (illustrative cost/value assumption tested against 4 scenarios)

XGBoost:

Scenario	Cost/Value Ratio	Optimal Threshold	% Clients Called	Subscribers Caught
Very cheap calls (\$2/\$80)	0.0250	0.05	94.5%	1047
Base assumption (\$5/\$80)	0.0625	0.31	44.5%	811
Moderate cost (\$10/\$80)	0.1250	0.51	17.8%	603
Expensive calls (\$20/\$80)	0.2500	0.63	11.6%	500

Random Forest:

Scenario	Optimal Threshold	% Clients Called	Subscribers Caught
Very cheap calls (\$2/\$80)	0.12	99.9%	1058
Base assumption (\$5/\$80)	0.35	48.9%	857
Moderate cost (\$10/\$80)	0.44	24.2%	658
Expensive calls (\$20/\$80)	0.57	10.0%	435

Finding: the optimal threshold moves substantially (0.05 to 0.63 for XGBoost) purely as a function of the assumed cost/value ratio — it is not a fixed property of either model. Real cost and value figures from the bank are required before a single deployment threshold can be finalized; this analysis provides the sensitivity range rather than a single number.

07. Cross-model profit comparison — discrepancy with PR-AUC

Model	Best Achievable Profit	Optimal Threshold	TP	FP	FN	Calls	% Called
XGBoost	\$44,750	0.31	811	3215	247	4026	44.5%
Random Forest	\$46,455	0.35	857	3564	201	4421	48.9%

Despite XGBoost's higher PR-AUC (0.4210 vs 0.4098), Random Forest achieves ~3.8% higher expected profit at its own optimal operating point. Flagged for statistical validation before acting on it.

08. Statistical validation (bootstrap, 500 resamples, seed 42)

Bootstrap resampling of test-set predictions (resampling only — no retraining) was used to check whether the profit gap was sampling noise:

- Mean (Random Forest – XGBoost) profit difference: **\$1,785**
- 95% confidence interval: [**\$282, \$3,465**] (excludes zero)
- Random Forest superior in **99.2%** of resamples

Conclusion: the profit advantage is real, not noise.

Limitation: this bootstrap captures test-set sampling variability only, not training variability (a different random seed or resampled training set could shift results); it should not be over-generalized beyond this specific train/test split.

09. Root-cause investigation — precision-recall crossover

Precision at matched recall levels was computed for both models to locate exactly where Random Forest overtakes XGBoost:

Recall	XGBoost Precision	Random Forest Precision
0.60	0.3376	0.3148
0.65	0.2794	0.2724
0.70	0.2331	0.2556
0.75	0.2066	0.2230
0.80	0.1733	0.1973
0.90	0.1440	0.1572

Crossover point (interpolated): **recall** ≈ 0.66 . Below this, XGBoost leads; above it, Random Forest leads by a widening margin. Both models' profit-optimal thresholds (XGBoost recall 76.7%, Random Forest recall 81.0%) sit past this crossover — directly explaining the profit reversal in section 7. Possible interpretation (not directly tested, stated as a hypothesis): XGBoost's boosting objective may sharpen confidence on the easiest, highest-signal cases at the cost of calibration further down the ranking, while Random Forest's bagging behaves more consistently across the full range.

10. Conclusion — model selection reversed

Final model: Random Forest, reversing the Model Comparison Log's initial XGBoost selection.

Justification: PR-AUC is a global metric averaged across all thresholds, while deployment only depends on model behavior at the specific, cost-justified operating point used — and Random Forest is statistically confirmed superior there (section 8). Recommended operating threshold ≈ 0.35 under the illustrative \$5/\$80 cost assumption, with the sensitivity range in section 6 provided for when real bank figures are available. This threshold-based call list must also be reconciled with the bank's actual call-center capacity — if capacity is more limited than the $\sim 48.9\%$ of clients this threshold implies the capacity constraint binds first and the ranked list (not the raw threshold) should govern who gets called.